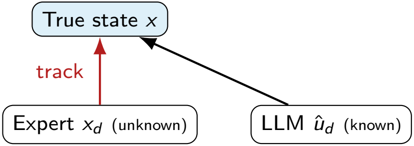
Certified Tracking of LLM-Predicted Trajectories

What Is Tracking What, Using What

Three trajectories in play:

- True state $x(t)$ — actual robot, known dynamics
- Desired/expert state $x_d(t)$ — from RRT/MPC, never observed at test time
- LLM prediction $\hat{u}_d$ — estimate of expert input u_d, computable at test time



Goal: design feedback so x provably tracks the *unobservable* x_d, using only $\hat{u}_d$.

Known Quantities

Design-time (always known)

- True dynamics structure: $\dot{x} = f(x, u)$ (or its linearization A, B)
- Freedom to design a feedback gain / metric (K, or $M(x)$, α)
- Calibration data $\{(x_{d,i}, u_{d,i}, \hat{u}_{d,i}, E_i)\}_{i=1}^{N}$, exchangeable with test conditions

Test-time (observed online)

- Environment E
- LLM prediction $\hat{u}_d = u_{vlm}(x, E)$
- True state $x(t)$ (measured)

Unknown Quantities & the Core Definitions

Unknown at test time

- Expert trajectory and control $x_d(t), u_d(t)$ — RRT/MPC is *not* re-run online
- Realized prediction gaps d — only its calibrated statistics are known

Tracking error:

$$e := x - x_d$$

Implementable controller:

$$u := -K\hat{e} + \hat{u}_d, \qquad \hat{e} := x - \hat{x}_d$$

Where the two tools split the work

Contraction/Lyapunov $\rightarrow$ guarantees e stays bounded, given $\|d\|$ is bounded. Conformal prediction $\rightarrow$ supplies that bound on $\|d\|$, from calibration data alone.

Error Dynamics With the Implementable Controller

Substitute $\hat{e} = x - \hat{x}_d$ **into** $e = x - x_d$:

$$\hat{e} = (x - x_d) + (x_d - \hat{x}_d) = e - d_x, \qquad d_x := \hat{x}_d - x_d$$

Error dynamics (with $u = -K\hat{e} + \hat{u}_d = -K(e - d_x) + \hat{u}_d$):

$$\dot{e} = Ae + B\big(-K(e - d_x) + \hat{u}_d\big) = (A - BK)e + \underbrace{B(\hat{u}_d - u_d)}_{d_u} + BKd_x$$

Two disturbance terms now, not one

$$\dot{e} = (A - BK)e + Bd_u + BKd_x$$

$d_u = \hat{u}_d - u_d$ (LLM control-prediction gap) and $d_x = \hat{x}_d - x_d$ (LLM state-prediction gap) – both unobservable at test time, both need conformal bounds.

Conformal Prediction: Two Separate Nonconformity Scores

Calibration data: N past triples, assumed exchangeable with the test environment: $\{(x_{d,i}, u_{d,i}, \hat{u}_{d,i}, E_i)\}_{i=1}^{N}$ **Score 1 – control-prediction gap:**

$$S_i^u := \sup_{t\in[0,T]} \left\| \hat{u}_d(x_{d,i}, E_i) - u_{d,i}(t) \right\|$$

Score 2 – state-prediction gap:

$$S_i^x := \sup_{t\in[0,T]} \left\| \hat{x}_d(E_i) - x_{d,i}(t) \right\|$$

$$q^u_{1-\delta_1} = \left\lceil (1-\delta_1)(N+1) \right\rceil\text{-th smallest of } S_1^u, \ldots, S_N^u$$
$$q^x_{1-\delta_2} = \left\lceil (1-\delta_2)(N+1) \right\rceil\text{-th smallest of } S_1^x, \ldots, S_N^x$$

Combining the Two Bounds: Boole's Inequality

Boole's inequality (union bound – holds for any two events, no independence needed):

$$\mathbb{P}[F_1 \cup F_2] \leq \mathbb{P}[F_1] + \mathbb{P}[F_2] \leq \delta_1 + \delta_2$$

Since $\mathbb{P}[F_1^c \cap F_2^c] = 1 - \mathbb{P}[F_1 \cup F_2]$:

Combined probabilistic guarantee

$$\mathbb{P}\Big[\, \|d_u\| \leq q^u_{1-\delta_1} \text{ and } \|d_x\| \leq q^x_{1-\delta_2} \,\Big] \geq 1 - \delta_1 - \delta_2 =: 1 - \delta$$

Standard choice: split the overall budget evenly, $\delta_1 = \delta_2 = \delta/2$.

The Resulting Certificate (Schematic)

$$\mathbb{P}\Big[\, \|x(t) - x_d(t)\| \le \rho(t)\ \forall t \in [0, T] \,\Big] \ge 1 - \delta$$

Contraction
(engineered: K, M, α)

Conformal Prediction
(statistical: $q_{1-\delta}$)

Certified probabilistic tube $\rho(t)$
around unobserved x_d

Next: instantiate this on a linear system (Lyapunov), then a nonlinear one (contraction).

Linear Dynamics Setup

True dynamics (known linear model):

$$\dot{x} = Ax + Bu$$

Expert/desired trajectory:

$$\dot{x_d} = Ax_d + Bu_d$$

Implementable controller (feedback on $\hat{e}$, not the unobservable e):

$$u = -K\hat{e} + \hat{u}_d, \qquad \hat{e} := x - \hat{x}_d, \qquad e := x - x_d$$

Error dynamics (derived earlier from $\hat{e} = e - d_x$):

$$\dot{e} = (A - BK)e + \underbrace{B(\hat{u}_d - u_d)}_{d_u} + \underbrace{BK(\hat{x}_d - x_d)}_{d_x}$$

Lyapunov Candidate & Contraction Condition

Let $P \succ 0$ and define

$$V(e) = e^\top P e$$

Design requirement (Lyapunov equation, guarantees $A - BK$ Hurwitz):

$$(A - BK)^\top P + P(A - BK) \preceq -2\alpha P, \qquad \alpha > 0$$

For linear systems, contraction is the Lyapunov condition for the error dynamics with $M = P$.

Differentiating V Along Trajectories

Product rule, then substitute $\dot{e} = (A - BK)e + Bd_u + BKd_x$ – two disturbance terms produce two cross terms:

$$\dot{V} = \dot{e}^\top Pe + e^\top P\dot{e} = e^\top \left[(A-BK)^\top P + P(A-BK)\right] e + 2e^\top PBd_u + 2e^\top PBKd_x$$

Using the design condition:

$$\dot{V} \leq -2\alpha V + 2e^\top PBd_u + 2e^\top PBKd_x$$

Next: bound each cross term separately using only $\|d_u\|, \|d_x\|$ and $\sqrt{V}$ – not their directions.

Bounding Both Cross Terms via Cauchy–Schwarz

$P = P^{1/2}P^{1/2}$, then Cauchy–Schwarz $p^\top q \le \|p\|\|q\|$, then bound each factor:

$$2e^\top PBd_u = 2(P^{1/2}e)^\top(P^{1/2}Bd_u) \le 2\|P^{1/2}e\|\,\|P^{1/2}Bd_u\|$$

$$2e^\top PBKd_x = 2(P^{1/2}e)^\top(P^{1/2}BKd_x) \le 2\|P^{1/2}e\|\,\|P^{1/2}BKd_x\|$$

Using $\|P^{1/2}e\| = \sqrt{V}$ and $\|P^{1/2}\| = \sqrt{\lambda_{max}(P)}$ in both:

Result – two bounded cross terms

$$2e^\top PBd_u \le 2\sqrt{\lambda_{max}(P)}\,\|B\|\,\|d_u\|\sqrt{V}$$
$$2e^\top PBKd_x \le 2\sqrt{\lambda_{max}(P)}\,\|BK\|\,\|d_x\|\sqrt{V}$$

Cross Terms, Conformally Bounded

With $\|d_u\| \leq q^u_{1-\delta_1}$ and $\|d_x\| \leq q^x_{1-\delta_2}$ simultaneously, w.p. $\geq 1-\delta$ (Boole's inequality, previous slides):

$$\dot{V} \leq -2\alpha V + 2\sqrt{\lambda_{max}(P)}\, c\, \sqrt{V}$$

$$c := \|B\| q^u_{1-\delta_1} + \|BK\| q^x_{1-\delta_2}$$

Same shape as the single-disturbance case – the two LLM-prediction gaps simply add inside the constant c, each weighted by how strongly it enters the dynamics (B vs. BK).

Comparison Lemma $\Rightarrow$ Tracking Tube

Let $W = \sqrt{V}$. Then $\dot{W} \leq -\alpha W + \frac{c}{2}$, giving

$$W(t) \leq W(0)e^{-\alpha t} + \frac{c}{2\alpha}(1 - e^{-\alpha t})$$

Since $\sqrt{\lambda_{min}(P)}\|e(t)\| \leq W(t)$:

$$\|e(t)\| \leq \sqrt{\frac{\lambda_{max}(P)}{\lambda_{min}(P)}}\|e(0)\|e^{-\alpha t} +$$

$$\frac{\sqrt{\lambda_{max}(P)}\left(\|B\|q^u_{1-\delta_1} + \|BK\|q^x_{1-\delta_2}\right)}{\alpha\sqrt{\lambda_{min}(P)}}(1 - e^{-\alpha t})$$

Probabilistic guarantee

$$\mathbb{P}\Big[\,\|x(t) - x_d(t)\| \leq \rho_{lin}(t)\ \forall t \in [0, T]\,\Big] \geq 1 - \delta$$

Exponential decay of initial error + a steady-state ball of radius $\propto (q^u_{1-\delta_1} + q^x_{1-\delta_2})/\alpha$ – without ever observing x_d or u_d, and using only the implementable controller.

True Dynamics: Double Integrator With Velocity Damping

State and input:

$$x = \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix} \in \mathbb{R}^4, \qquad u = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \in \mathbb{R}^2$$

Dynamics:

$$\dot{x} = \begin{bmatrix} v_x \\ v_y \\ -1.1v_x + 1.1a_x \\ -1.1v_y + 1.1a_y \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1.1 & 0 \\ 0 & 0 & 0 & -1.1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1.1 & 0 \\ 0 & 1.1 \end{bmatrix}$$

Assumption: PX4 Inner Loop as an Idealized Tracking Oracle

What PX4's cascaded PID actually controls

Position → velocity → attitude → body-rate loops, converting a commanded $(p_{ref}, v_{ref}, a_{ref})$ into thrust T and torques τ, then into motor commands.

Idealization assumed here

The entire chain $(p_{ref}, v_{ref}, a_{ref}) \to (T, \tau) \to \ddot{p}_{actual}$ is assumed *instantaneous and exact*:

$$\ddot{p}_{actual}(t) \equiv a_{ref}(t)$$

i.e. PX4 perfectly and immediately realizes any commanded acceleration. Attitude, angular rate, and the rotation matrix R do not appear in x – they are not modeled, only assumed tracked.

Consequence

Robot is treated as a point mass; $u = a$ is the design-level control input; the true rigid-body torque/thrust never appears in this formulation.

Setpoint Pipeline: Design-Level u to PX4

$$u(t) = a_{ref}(t) \;\longrightarrow\; \big(p_{ref}(t), v_{ref}(t), a_{ref}(t)\big)$$

$$\longrightarrow\; \texttt{TrajectorySetpoint} \;\longrightarrow\; \text{PX4 cascade}$$

$$p_{ref}(t) = \int_0^t v_{ref}(s)\, ds, \qquad v_{ref}(t) = \int_0^t a_{ref}(s)\, ds$$

Online: integrate the designed $u(t) = a_{ref}(t)$ forward to obtain consistent position/velocity setpoints before publishing.

Expert (x_d) and LLM ($\hat{x}_d$) Trajectories: Shared Pipeline

Assumption: LLM waypoints are refined and verified before use

Raw LLM output $\{\hat{P}_i\}$ is passed through a **refinement and verification pipeline** (collision/feasibility check, velocity/acceleration-limit clipping, reordering/pruning) before being treated as usable waypoints – raw, unverified LLM output is never interpolated directly.

Shared downstream pipeline (RRT waypoints identical procedure)

$$\{P_i\} \text{ (or } \{\hat{P}_i\}) \xrightarrow{\text{Bézier fit}} B(\tau) \xrightarrow{\text{differential flatness}} p(t)$$

$$v(t) = \dot{p}(t), \qquad a(t) = \ddot{p}(t)$$

Resulting trajectories live in the same state space as x

$$x_d(t) = (p_d, v_d)(t), \qquad \hat{x}_d(t) = (\hat{p}_d, \hat{v}_d)(t)$$

$$u_d = a_d = \dot{v}_d, \qquad \hat{u}_d = \hat{a}_d = \dot{\hat{v}}_d$$

Numeric CARE Solution for This A, B

$$Q = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 1 \end{bmatrix} \qquad \text{(per-axis, e.g. the } x\text{-channel)}$$

Solving $A^\top P + PA - PBR^{-1}B^\top P + Q = 0$:

$$P = \begin{bmatrix} 8.803 & 2.875 \\ 2.875 & 1.622 \end{bmatrix}, \qquad K = \begin{bmatrix} 3.162 & 1.784 \end{bmatrix}$$

$$A - BK = \begin{bmatrix} 0 & 1 \\ -3.479 & -3.062 \end{bmatrix}, \qquad \mathrm{eig}(A - BK) = -1.531 \pm 1.065j$$

$$\lambda_{min}(P) = 0.613, \qquad \lambda_{max}(P) = 9.812, \qquad \alpha = 1.531$$

Resulting Tube, This Numeric Instance

$$\|e(t)\| \le \sqrt{\tfrac{9.812}{0.613}}\,\|e(0)\|\,e^{-1.531t} +$$

$$\frac{\sqrt{9.812}\big(\|B\|q^{u}_{1-\delta_1} + \|BK\|q^{x}_{1-\delta_2}\big)}{1.531\sqrt{0.613}}\big(1 - e^{-1.531t}\big)$$

$$\sqrt{\tfrac{9.812}{0.613}} = 4.005, \qquad \frac{\sqrt{9.812}}{1.531\sqrt{0.613}} = 2.638$$

$$\|e(t)\| \le 4.005\,\|e(0)\|e^{-1.531t} +$$

$$2.638\big(1.1\,q^{u}_{1-\delta_1} + 1.96\,q^{x}_{1-\delta_2}\big)\big(1 - e^{-1.531t}\big)$$

The Nonlinear System Being Linearized

Full nonlinear rigid-body dynamics $\dot{x} = f(x, u)$,
$x = (p, v, \phi, \theta, \psi, \omega) \in \mathbb{R}^{12}$, $u = (T, \tau_x, \tau_y, \tau_z)$:

$$\dot{p} = v, \qquad \ddot{z} = g - (\cos\phi\cos\theta)\frac{T}{m}$$

$$\ddot{\phi} = \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) - \frac{I_R}{I_x}\dot{\theta}\, g(\mathbf{u}) + \frac{L}{I_x}\tau_x$$

$$\ddot{\theta} = \dot{\phi}\dot{\psi}\left(\frac{I_z - I_x}{I_y}\right) + \frac{I_R}{I_y}\dot{\phi}\, g(\mathbf{u}) + \frac{L}{I_y}\tau_y, \qquad \ddot{\psi} = \dot{\phi}\dot{\theta}\left(\frac{I_x - I_y}{I_z}\right) + \frac{1}{I_z}\tau_z$$

This is the object Part B approximates – nothing here is linear

$\ddot{\phi}, \ddot{\theta}, \ddot{\psi}$ contain products of rates ($\dot{\theta}\dot{\psi}$, $\dot{\phi}\dot{\psi}$, $\dot{\phi}\dot{\theta}$) and trigonometric attitude coupling – exactly the terms a first-order Taylor expansion at hover discards. The next slides linearize *this* f, not a simplified stand-in for it.

Linearization Setup

Operating point (hover):

$$x^* = (p^*, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \qquad u^* = (mg, 0, 0, 0)$$

Taylor expansion, $\delta x = x - x^*$, $\delta u = u - u^*$:

$$\dot{\delta x} \approx \underbrace{\frac{\partial f}{\partial x}\bigg|_{x^*,u^*}}_{A} \delta x + \underbrace{\frac{\partial f}{\partial u}\bigg|_{x^*,u^*}}_{B} \delta u + \rho(\delta x)$$

Assumption: Bound on the Taylor Remainder

What is discarded

$[\omega]_\times$, $\omega \times J\omega$ (zero at $\omega = 0$, growing quadratically away from it); $\sin\theta \approx \theta$ truncation (remainder $O(\theta^3)$); off-diagonal attitude/thrust coupling beyond first order.

Assumed Lipschitz-type bound, valid on a candidate region $\mathcal{B}_r = \{\|\delta x\| \le r\}$

$$\|\rho(\delta x)\| \le L\|\delta x\|^2, \qquad L = 3.5 \ \text{(dummy, airframe-dependent)}$$

This bound is in addition to d_x, d_u

ρ is a *model-fidelity* error (true f vs. its linearization) – distinct from $d_x = \hat{x}_d - x_d$, $d_u = \hat{u}_d - u_d$ (LLM-prediction errors). Three error sources now stack, not two.

State and Input

$$x = (p, v, \phi, \theta, \psi, \omega) \in \mathbb{R}^{12}, \qquad u = (T, \tau_x, \tau_y, \tau_z) \in \mathbb{R}^4$$

$$m = 1.5\,\mathrm{kg}, \quad g = 9.81\,\mathrm{m/s^2}, \quad J = \mathrm{diag}(0.0347, 0.0458, 0.0977)\,\mathrm{kg\,m^2}$$

Explicit A Matrix, Block Form (12×12)

Ordering $x = (p_x, p_y, p_z, v_x, v_y, v_z, \phi, \theta, \psi, \omega_x, \omega_y, \omega_z)$:

$$A = \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & g & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -g & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Only the (v, Θ) block is nonzero beyond the kinematic identities – this is exactly the $g\theta, -g\phi$ small-angle coupling, the *only* place gravity enters the linearized model.

Explicit B Matrix (12×4)

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{I_x} & 0 & 0 \\ 0 & 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & 0 & \frac{1}{I_z} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.667 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 28.8 & 0 & 0 \\ 0 & 0 & 21.8 & 0 \\ 0 & 0 & 0 & 10.2 \end{bmatrix}$$

Lyapunov-Satisfaction Error Under the True Nonlinear Remainder

Design condition only certified for the linear part

$$(A - BK)^\top P + P(A - BK) \preceq -2\alpha P$$

Actual $\dot{V}$ picks up an extra, uncertified term

$$\dot{V} = e^\top \left[(A - BK)^\top P + P(A - BK)\right] e + 2e^\top P\rho(\delta x)$$
$$\leq -2\alpha V + 2\sqrt{\lambda_{max}(P)}\, L\|\delta x\|^2 \sqrt{V}$$

Region of validity r^* (Lyapunov-remainder bound, certified but conservative)

$$r^* = \frac{\lambda_{min}(Q)}{2L\,\lambda_{max}(P)}, \qquad L = 3.5,\ \lambda_{min}(Q) = 0.1,\ \lambda_{max}(P) = 10.31$$

$$r^* \approx 0.00139 \quad \text{(rad-scale state error, conservative)}$$

Designing $K(t)$ via TVLQR

$$-\dot{P}(t) = A(t)^\top P(t) + P(t)A(t) - P(t)B(t)R^{-1}B(t)^\top P(t) + Q$$

$$P(t_f) = P_f, \qquad K(t) = R^{-1}B(t)^\top P(t)$$

$$Q = \mathrm{diag}(10, 10, 10, 1, 1, 1, 5, 5, 1, 0.1, 0.1, 0.1)$$

$$R = \mathrm{diag}(0.5, 1, 1, 1)$$

Steady-state numeric check at one trajectory point (algebraic CARE):

$$\max_i \mathrm{Re}\big(\mathrm{eig}(A - BK)\big) = -1.309$$

$$\lambda_{min}(P) = 0.0122, \quad \lambda_{max}(P) = 10.31$$

Expert (x_d) and LLM ($\hat{x}_d$): Min-Snap Optimization

Waypoints enter as constraints, not interpolation knots

$$\min_{p(t)} \int_0^T \|p^{(4)}(t)\|^2 \, dt$$

$$\text{s.t.} \quad p(t_i) = P_i, \quad \dot{p}(t_i^-) = \dot{p}(t_i^+), \quad \ddot{p}(t_i^-) = \ddot{p}(t_i^+)$$

Same problem, LLM waypoints (post refinement/verification)

$$\min_{\hat{p}(t)} \int_0^T \|\hat{p}^{(4)}(t)\|^2 \, dt \quad \text{s.t.} \quad \hat{p}(t_i) = \hat{P}_i$$

(same continuity constraints)

$$x_d = (p_d, v_d, \dots), \quad u_d = \text{flatness map}(p_d, \dots, p_d^{(4)})$$

$\hat{x}_d, \hat{u}_d$ identical, with $\hat{P}_i$ in place of P_i

Resulting Statistical Bound, Linearized Case

$$\|e(t)\| \leq \sqrt{\frac{\lambda_{max}(P)}{\lambda_{min}(P)}}\|e(0)\|e^{-\alpha t} +$$

$$\frac{\sqrt{\lambda_{max}(P)}\left(\|B\|q^{u}_{1-\delta_1} + \|BK\|q^{x}_{1-\delta_2}\right)}{\alpha\sqrt{\lambda_{min}(P)}}\left(1 - e^{-\alpha t}\right)$$

valid only for $\|e(t)\| \leq r^*$

(else $\rho(\delta x)$ term dominates, bound not certified)

Why a Constant K Cannot Work Here

The Jacobian itself depends on the state

$$\frac{\partial f}{\partial x}(x) \text{ contains } [\omega]_\times,\ \omega \times J\omega,\ \frac{\partial (R\hat{z}_B)}{\partial R}$$

– all state-dependent, none vanishing away from hover

Consequence

A constant K makes $A_{cl}(x) = \frac{\partial f}{\partial x}(x) + \frac{\partial f}{\partial u}(x)K$ – still a function of x. Hurwitz at one operating point does not imply Hurwitz at another; one gain cannot stabilize a linearization that itself moves with x.

How $K(x)$ Restores Contraction

Let the gain track the Jacobian

Design a *function* $K(x)$, not a constant, so that $A_{cl}(x) = \frac{\partial f}{\partial x}(x) + \frac{\partial f}{\partial u}(x)K(x)$ is well-behaved **at every** $x \in \mathcal{X}$.

What "well-behaved" must mean – accounting for $\dot{M}$, not a frozen snapshot

$$\dot{M}(x) + M(x)A_{cl}(x) + A_{cl}(x)^\top M(x) \preceq -2\alpha M(x), \qquad \forall x \in \mathcal{X}$$

$\dot{M} = \sum_i \frac{\partial M}{\partial x_i}\dot{x}_i$ is the *total* derivative along the actual flow – this is what lets the LMI certify the true time-varying system, not a sequence of disconnected frozen-point checks.

From LMI Feasibility to Contraction – the Mechanism

Virtual displacement along the flow

$$\dot{\delta x} = \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial u}\delta u, \qquad \delta u = K(x)\delta x$$

Once $M(x), K(x)$ solve the LMI everywhere on $\mathcal{X}$

For *any* two trajectories $x_1(t), x_2(t)$ of the closed-loop f remaining in $\mathcal{X}$:

$$\dot{V} \le -2\alpha V \implies d_M\big(x_1(t), x_2(t)\big) \le e^{-\alpha t} d_M\big(x_1(0), x_2(0)\big)$$

Convergence is produced, not assumed

If no (M, K) satisfying the LMI exists over $\mathcal{X}$, no convergence claim can be made – the synthesis step *is* the content of the guarantee.

Computing $K(x)$: Sum-of-Squares Synthesis

Change of variables (convexifying substitution):

$$W(x) := M(x)^{-1}, \qquad Y(x) := K(x)W(x)$$

LMI becomes affine in (W, Y):

$$\dot{W} - \left(\frac{\partial f}{\partial x} W + W \frac{\partial f}{\partial x}^\top\right) - \left(\frac{\partial f}{\partial u} Y + Y^\top \frac{\partial f}{\partial u}^\top\right) \succeq 2\alpha W, \quad \forall x \in \mathcal{X}$$

Positivstellensatz (region $\mathcal{X} = \{g_j(x) \geq 0\}$):

$$\text{LHS} - 2\alpha W = S_0(x) + \sum_j g_j(x) S_j(x), \qquad S_0, S_j \text{ SOS}$$

Solved via: polynomial parameterization of $W(x), Y(x)$ in monomials of x, SOS constraints $\rightarrow$ PSD matrix constraints, solved as an SDP using `SOSTOOLS`, `YALMIP`, or `cvxpy`+`MOSEK`/`SCS`.

Recovering $K(x)$ From the Solver Output

$$K(x) = Y(x)\,W(x)^{-1} = Y(x)\,M(x)$$

Illustrative reduced-order result (roll-rate subsystem, $\xi = (\phi, \dot{\phi})$, K acting on $\dot{\phi}$ only):

$$K(\dot{\phi}) = k_0 + k_1\dot{\phi}^2, \qquad k_0 = 4.0, \quad k_1 = 2.0$$

$$A_{cl}(\dot{\phi}) = -k_0 - k_1\dot{\phi}^2 + \underbrace{0.5\dot{\phi}^2}_{\text{gyroscopic-type growth}} = -4.0 - 1.5\dot{\phi}^2$$

Designing $M(x)$ for This $K(x)$

CCM condition, evaluated as an SOS certificate:

$$p(\dot{\phi}) := -2m_0 A_{cl}(\dot{\phi}) - 2\alpha m_0 \geq 0 \quad \forall \dot{\phi}$$

$$m_0 = 1.0, \quad \alpha = 3.0 \quad \Longrightarrow \quad p(\dot{\phi}) = 3.0\,\dot{\phi}^2 + 2.0$$

SOS certificate (globally, no region restriction needed here)

$$p(\dot{\phi}) = \left(\sqrt{3}\,\dot{\phi}\right)^2 + \left(\sqrt{2}\right)^2 \quad \Longrightarrow \quad p(\dot{\phi}) \geq 0 \;\; \forall \dot{\phi} \in \mathbb{R}$$

$M = m_0 = 1.0$ (constant suffices in this slice), $\quad \lambda_{min}(M) = \lambda_{max}(M) = 1.0$

$A_{cl}(0) = -4.0 \leq -\alpha = -3.0$ ✓, $\quad A_{cl}(\dot{\phi})$ only more negative as $|\dot{\phi}|$ grows

Full Nonlinear Model (Restated From the Rigid-Body Equations)

$$\ddot{x} = -(\cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi) \cdot \frac{u_1}{m}$$

$$\ddot{y} = -(\cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi) \cdot \frac{u_1}{m}$$

$$\ddot{z} = g - (\cos\phi \cos\theta) \cdot \frac{u_1}{m}$$

$$\ddot{\phi} = \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) - \frac{I_R}{I_x}\dot{\theta}\, g(\mathbf{u}) + \frac{L}{I_x} u_2$$

$$\ddot{\theta} = \dot{\phi}\dot{\psi}\left(\frac{I_z - I_x}{I_y}\right) + \frac{I_R}{I_y}\dot{\phi}\, g(\mathbf{u}) + \frac{L}{I_y} u_3, \qquad \ddot{\psi} = \dot{\phi}\dot{\theta}\left(\frac{I_x - I_y}{I_z}\right) + \frac{1}{I_z} u_4$$

Expert (x_d) and LLM ($\hat{x}_d$): Min-Snap, Same Constraint Structure

$$\min_{p(t)} \int_0^T \|p^{(4)}(t)\|^2 dt \quad \text{s.t.} \quad p(t_i) = P_i, \quad (C^2 \text{ continuity})$$

$$x_d = \Phi^{-1}\big(p_d, \dot{p}_d, \ddot{p}_d, \dddot{p}_d, p_d^{(4)}\big)$$

(differential flatness, full $SE(3)$ state)

Identical problem with $\hat{P}_i$ (post refinement/verification) gives $\hat{x}_d, \hat{u}_d$.

Contraction Within Conformal-Probabilistic Bounds

Riemannian energy and disturbance-driven rate (mirrors the linear case, $\sqrt{V} \to d_M$):

$$\dot{E}(\gamma(\cdot,t)) \leq -2\alpha E(\gamma(\cdot,t)) + 2\sqrt{m}\, d_M(x, x_d)\, \|d\|$$

$$d = \hat{u}_d - u_d \quad \text{(or state-gap, via Boole's inequality as in Parts A/B)}$$

Comparison lemma, same machinery, $\sqrt{V} \to d_M$:

$$d_M\big(x(t), x_d(t)\big) \leq d_M\big(x(0), x_d(0)\big)\, e^{-\alpha t} + \frac{\sqrt{m}\, q_{1-\delta}}{\alpha}\big(1 - e^{-\alpha t}\big)$$

Probabilistic guarantee, full nonlinear case

$$\mathbb{P}\Big[\, d_M\big(x(t), x_d(t)\big) \leq \rho_{CCM}(t)\ \forall t \in [0, T] \Big] \geq 1 - \delta, \quad \forall x \in \mathcal{X}$$

What Is Computed Online vs. Offline, Per Method

Double Integrator

Offline: A, B fixed; solve CARE once for P, K; calibrate $q^u_{1-\delta_1}, q^x_{1-\delta_2}$.
Online: measure $x(t)$, compute $\hat{e} = x - \hat{x}_d$, $u = -K\hat{e} + \hat{u}_d$; integrate u for p_{ref}, v_{ref}; publish `TrajectorySetpoint`.

Linearized Quadrotor (TVLQR)

Offline: linearize f along (x_d, u_d); solve Riccati ODE backward for $P(t), K(t)$; bound Taylor remainder L, compute r^*; calibrate q^u, q^x.
Online: measure $x(t)$; look up $K(t)$ at current time; $u = -K(t)\hat{e} + \hat{u}_d$; monitor $\|\hat{e}\|$ against r^*.

Full Nonlinear (CCM)

Offline: SOS/SDP synthesis of $W(x), Y(x)$ over $\mathcal{X}$ (solver: SOSTOOLS/YALMIP/cvxpy); recover $M(x), K(x)$; calibrate $q_{1-\delta}$.
Online: measure $x(t)$; evaluate $K(x(t))$ (closed-form/lookup); $u = \hat{u}_d + K(x)(x - \hat{x}_d)$; geodesic distance d_M tracked implicitly via the certified bound, not recomputed online.